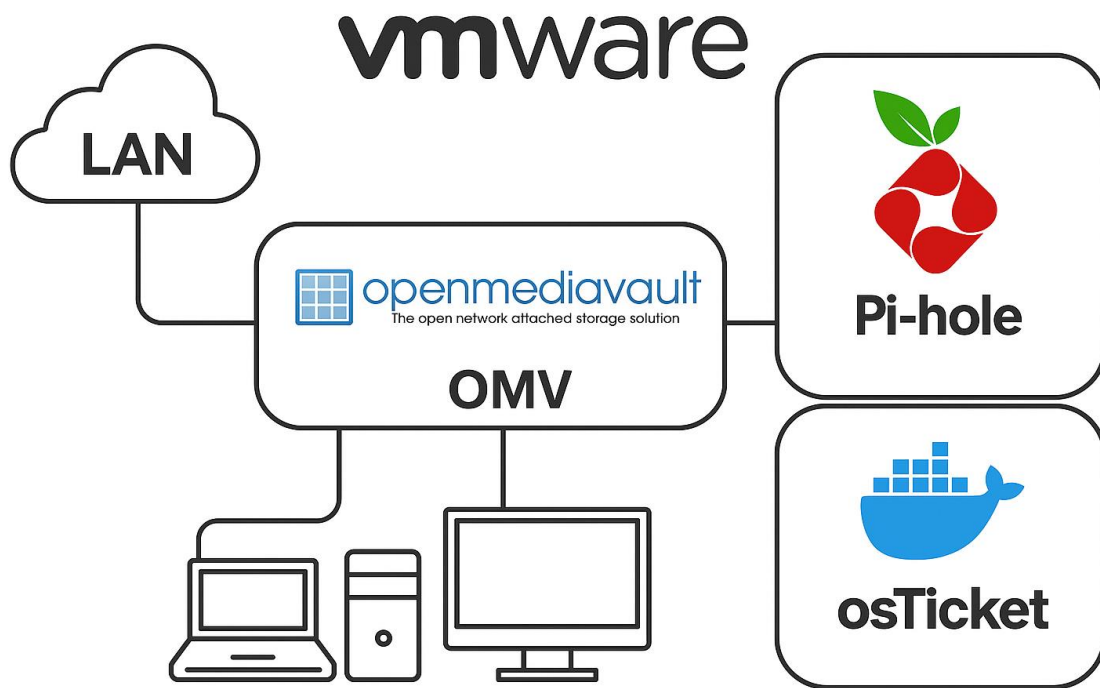


Introduction to infrastructures

Assignments Mac Appendices 2025 - 2026



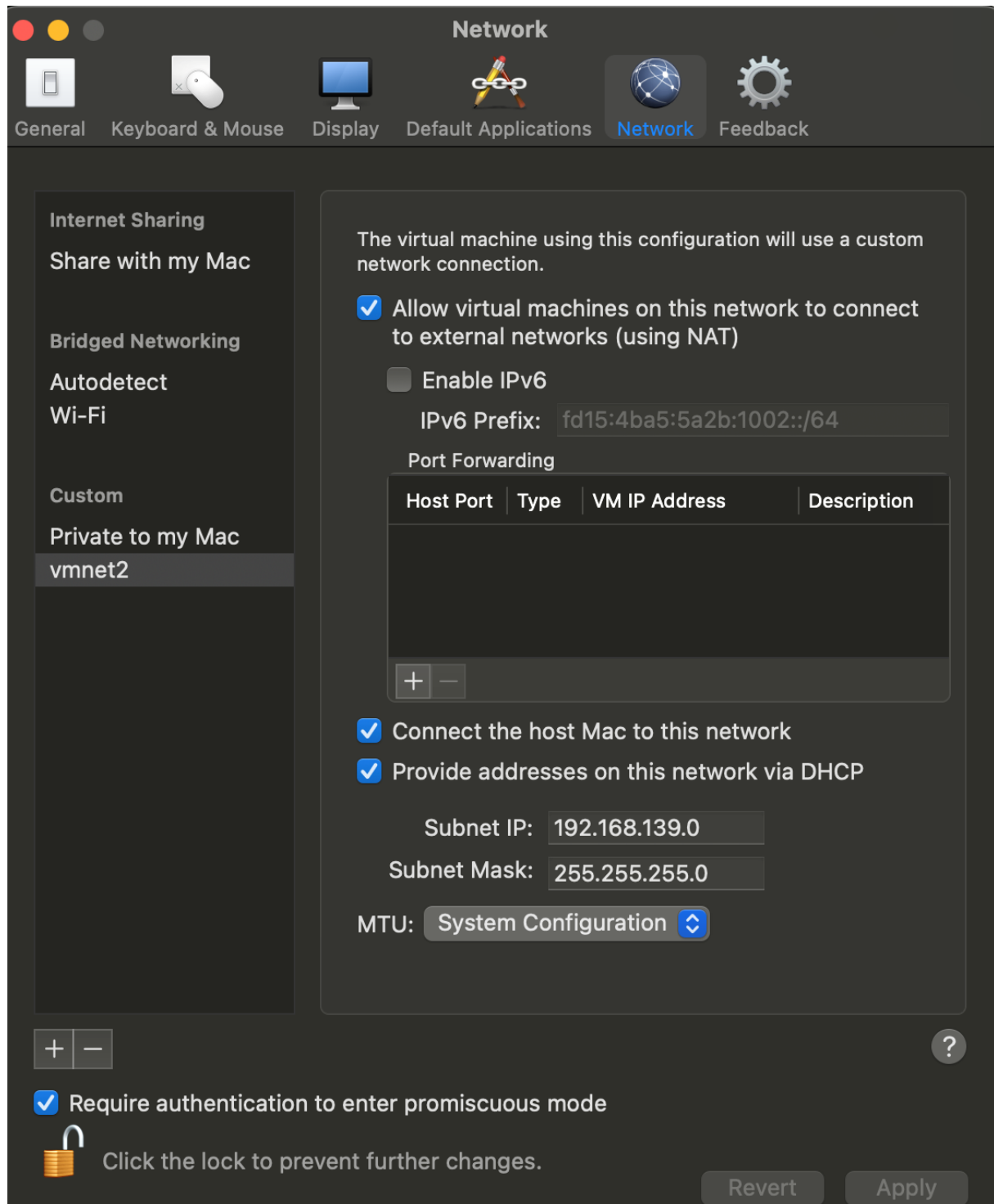
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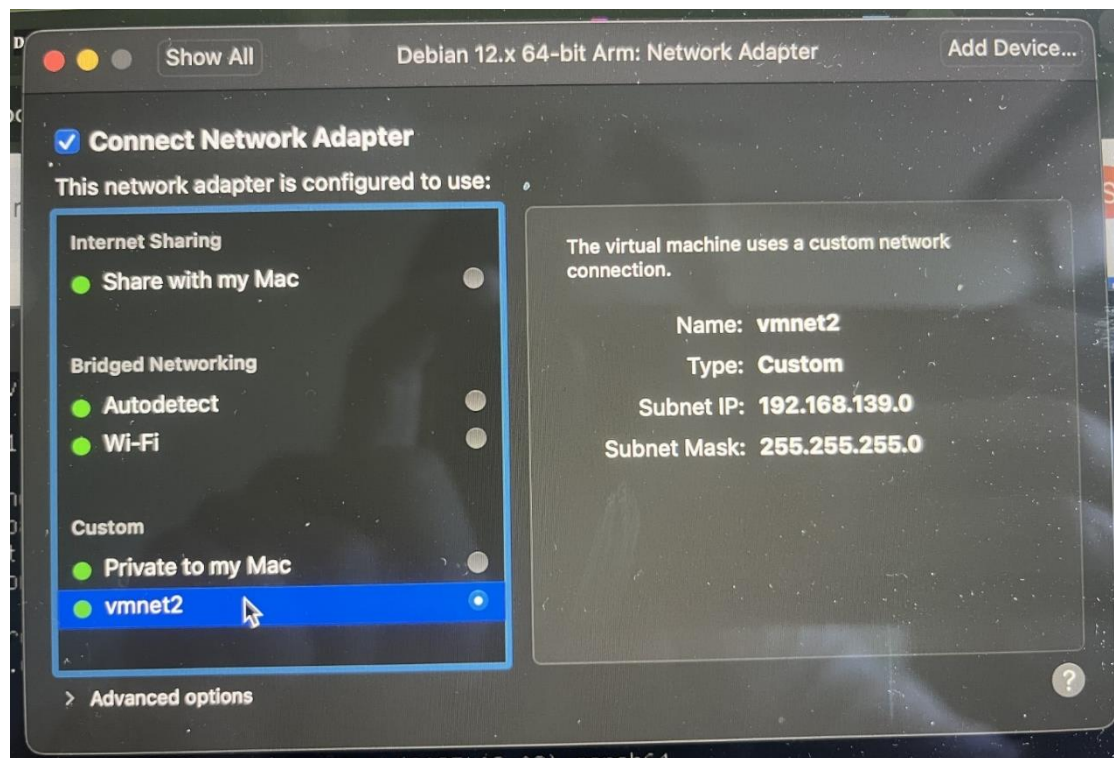
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Week 1 - Mac

1.1 VMware Fusion Pro network settings





Screenshots provided by: Sofiia Ivanova

Week 2 - Mac

No differences

Week 3 - Mac

3.3 Installing osTicket app via docker compose

osTicket docker compose yaml file (**Testing and debugging done by:** David Denis):

```
---
services:
  osticket:
    image: tiredofit/osticket:3.5.2
    container_name: OsTicket
    depends_on:
      - osticket_db
    ports:
      - "8080:80" # Maps host port 8080 to container port 80
    networks:
      backend-network:
    environment:
      # Database configuration
      DB_HOST: osticket_db
      DB_PORT: 3306
      DB_NAME: my_database_name
      DB_USER: some_user
      DB_PASS: some_password
      # Admin configuration
      ADMIN_USER: mike
      ADMIN_PASS: Welkom#01
      ADMIN_EMAIL: yourown@email
      ADMIN_FIRSTNAME: Mike
      ADMIN_LASTNAME: Sierra
      # Installation config
      INSTALL_SECRET: mysecret
      INSTALL_URL: http://osticket.local
      INSTALL_NAME: Helpdesk
      CRON_INTERVAL: 1
    volumes:
      - CHANGE_TO_COMPOSE_DATA_PATH/osticket/config:/var/www/html
    restart: unless-stopped

  osticket_db:
    image: mariadb:11.3-jammy
    container_name: OsTicket_DB
    security_opt:
      - no-new-privileges:true
    environment:
      MYSQL_ROOT_PASSWORD: root_password
      MYSQL_USER: some_user
      MYSQL_PASSWORD: some_password
      MYSQL_DATABASE: my_database_name
    volumes:
      - CHANGE_TO_COMPOSE_DATA_PATH/osticket/db:/var/lib/mysql:rw
    networks:
      - backend-network
    restart: unless-stopped

networks:
  backend-network:
    name: osticket-backend-network
```

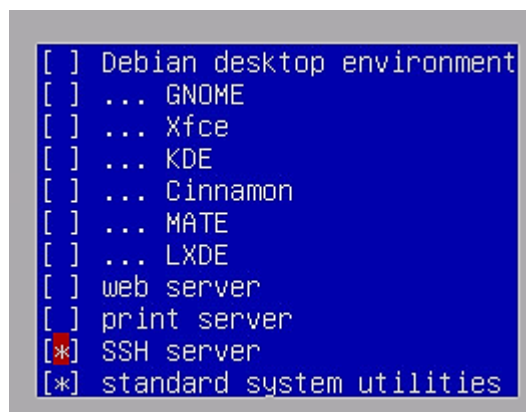
See week 3 in the intro infra [app](#) for the docker compose yaml file. The IP of Pi-hole is now: 192.168.139.11:8080. Same IP as the OpenMediaVault server but on another port.

Week 4 – Mac

On a Mac we are not going to use the docker container for Pi-hole. We need to create a new virtual machine and install Pi-hole on that.

4.1 Debian 12 Server in VMware for Pi-hole

- Install a new Debian 12 Server VM
- 1cpu, 2cores, 2GB ram, 32GB hard disk, NAT network adapter, CD/DVD drive.
- ISO: [AMD64](#), [ARM64](#)
- See installation video in the intro infra [app](#) at week 1.
- Use hostname: pihole<your student number>
- Domain: local
- Don't create a root user, create a user with sudo privileges that has your first name.
- Don't install a Graphical desktop environment



Select SSH server and standard system utilities only!

Finish the Debian 12 installation and let it reboot.

4.2 Install Pi-hole on Debian 12 Pi-hole VM

Step 1: Update your system

Use the following terminal commands:

```
sudo apt update && sudo apt upgrade -y
sudo apt install curl sudo -y
```

Step 2: Identify your network interface

Use the following terminal command:

```
ip a
```

Write down the actual network interface name. For example: ens33.

Step 3: Configure a static IP using systemd-networkd

Create a new network config file for your interface:

```
sudo nano /etc/systemd/network/20-static.network
```

Example for interface eth0, paste the following in the configuration file:

```
[Match]
# Change this to match your network interface name (ens18, enp0s3, etc.)
Name=eth0

[Network]
Address=192.168.139.3/24
Gateway=192.168.139.2
DNS=192.168.139.2
```

Enable and restart systemd-networkd:

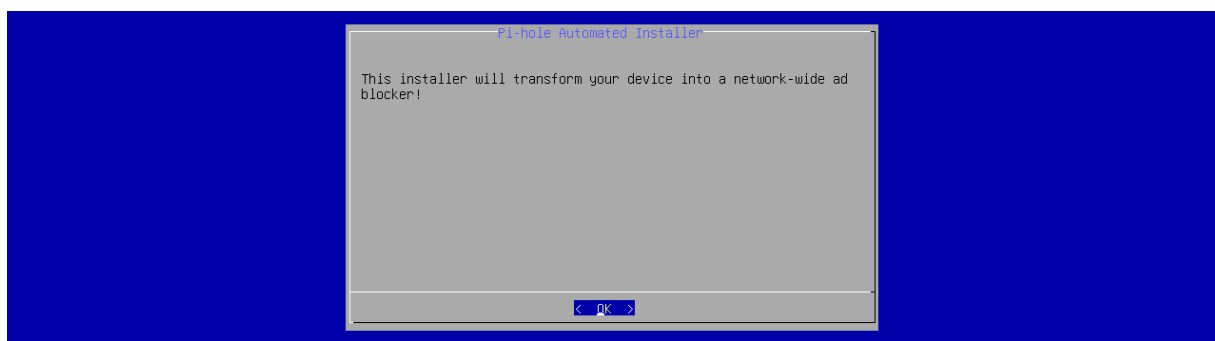
Use the following terminal commands:

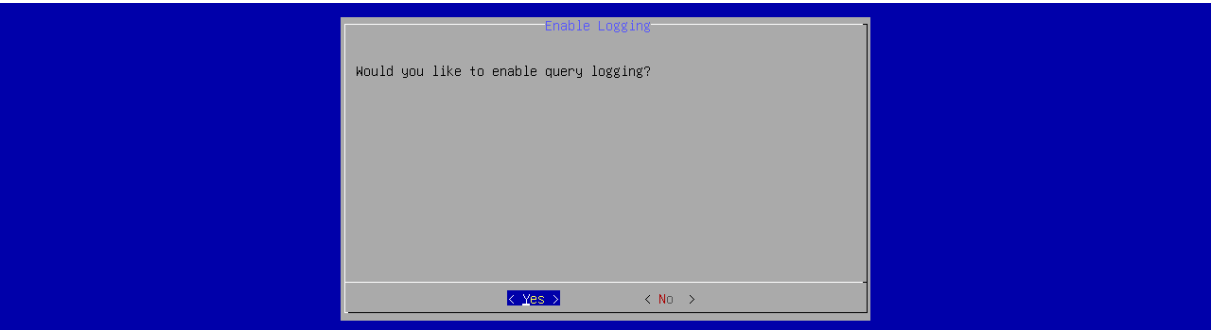
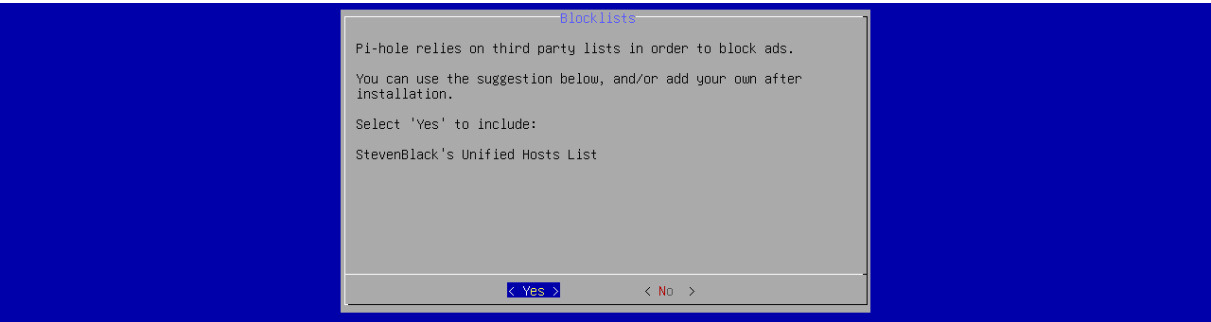
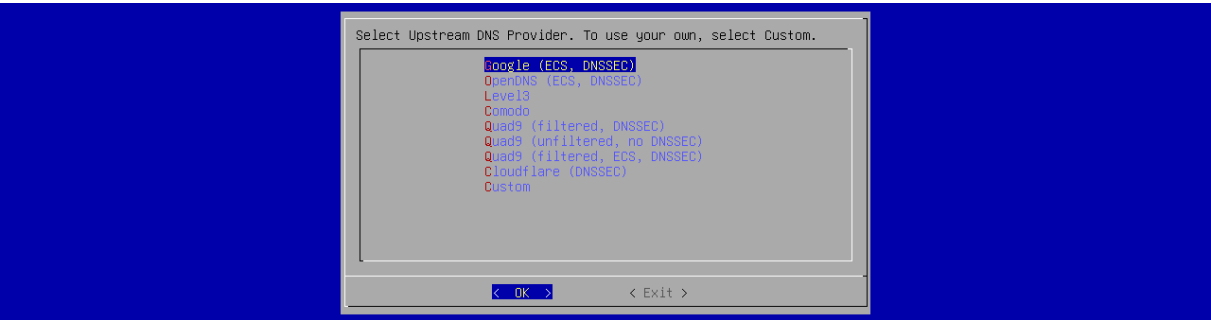
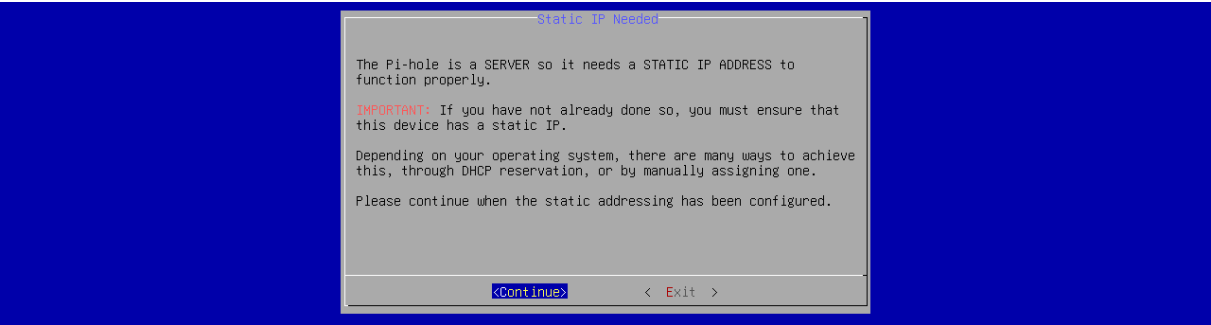
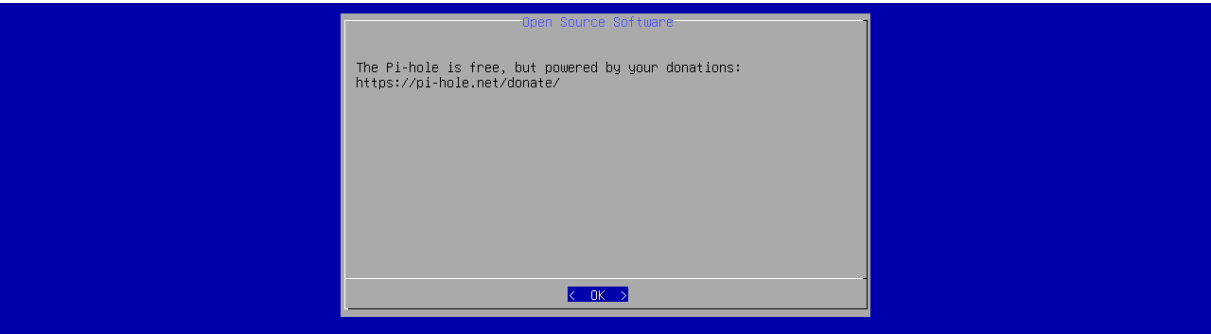
```
sudo systemctl enable systemd-networkd
sudo systemctl restart systemd-networkd
```

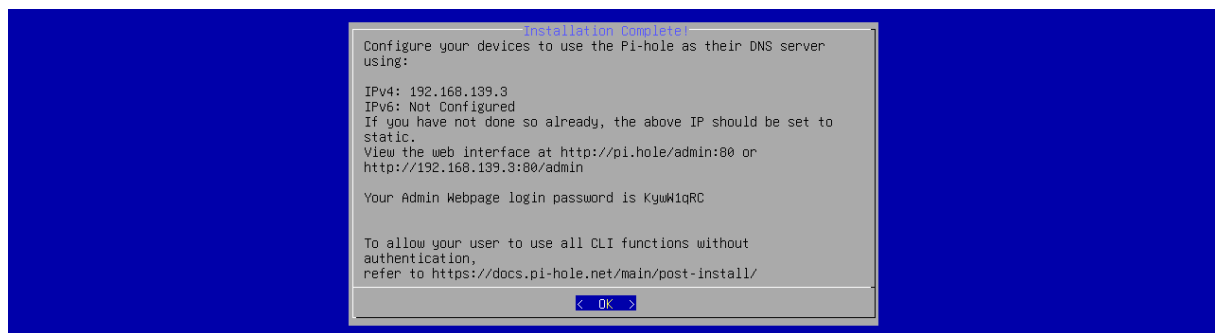
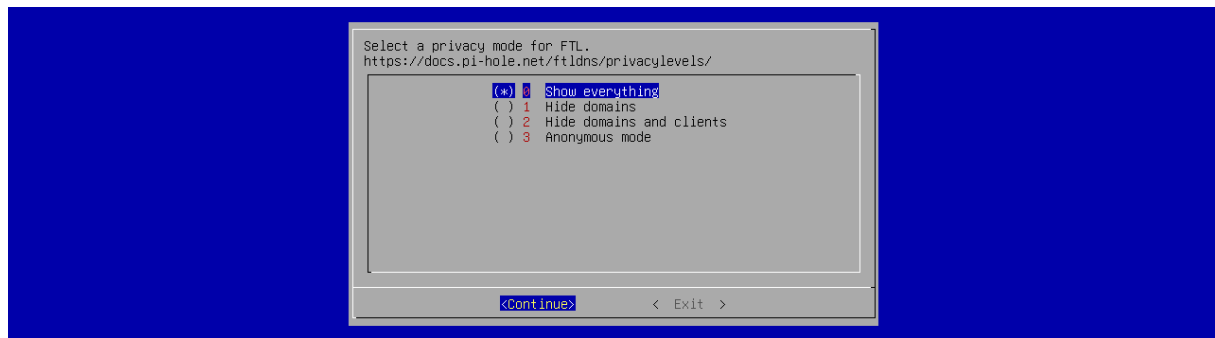
Step 4: Install Pi-hole

Use the following terminal command:

```
curl -sSL https://install.pi-hole.net | sudo bash
```







Step 5: Verify Pi-hole

Use the following terminal command:

```
sudo pihole status
```

Step 6: Set admin password

Use the following terminal command:

```
sudo pihole setpassword 'Welkom#01'
```

Step 7: Configure Pi-hole and LAN clients

See assignments document chapter 4 paragraphs 4.3 and 4.4

Week 5 - Mac

No differences

Week 6 - Mac

No differences